Worksheet 13

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already established

1. A battery manufacturer claims their product has a life expectancy of 90 hours. An improvement production process is believed to make an increase in the life expectancy of batteries. A sample of 36 batteries showed an average life of 93 hours. Assume from past information, the standard deviation of the life expectancy is 9 hours.

   a. Formulate the hypotheses for this problem.

   $H_0: \mu \leq 90$

   $H_a: \mu > 90$

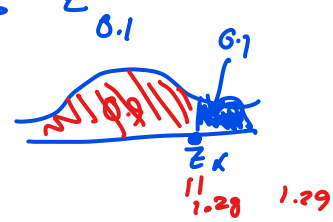
   b. Calculate the test statistic.

   c. Make a decision at 0.1 significance level. State your conclusion in terms of the problem.

   d. Make a decision at 0.01 significance level. What is your conclusion.

(b) $$Z_{\bar{x}} = \frac{\bar{x} - \mu_{H_0}}{\sigma/\sqrt{n}} = \frac{\overset{3}{93 - 90}}{9/\underset{6}{\sqrt{36}}} = 2$$

(c) $Z_\alpha \rightarrow Z_{0.1}$



$\alpha \rightarrow Z_{0.01}$

$2.32 < Z_{0.01} \approx 2.33$

$\underset{2}{Z_{\bar{x}}} > \underset{128..}{Z_\alpha} \xrightarrow{\text{reject}}$ Reject the null hypothesis that average life expectancy is 90 at $\alpha = 0.1$

$\underset{2}{Z_{\bar{x}}} \overset{?}{>} \underset{2.33}{Z_\alpha} \xrightarrow{\text{No}}$ We can't reject at $\alpha = 0.01$

2.32Y

P-value $= P(Z \geq \underset{2}{Z_{\bar{x}}}) = 1 - P(Z < 2) = 1 - 0.9772 = \boxed{0.0228}$

$0.0228 < 0.1$

$\boxed{\text{P-value} < \alpha}$ → Reject

α

Part c

$\alpha = 0.1 \rightarrow$ P-val $< \alpha \rightarrow$ Reject H_0

Part d

$\alpha = 0.01 \rightarrow$ P-val $> \alpha \rightarrow$ we fail to reject.

$0.0228 > 0.01$

2. The average gasoline price of one of the major oil companies is \$1.75 per gallon. Because of cost reduction measures, it is believed that there has been a significant reduction in the average price. In order to test this belief, a sample of 36 of the company's gas stations were randomly selected and yielded an average price \$1.65 per gallon. Assume the standard deviation of the population is \$0.12.

✓ a. State the null and the alternative hypothesis for this problem.

b. Calculate the test statistic.

c. At 5% level of significance, test the company's claim.

$$\begin{cases} H_0: \mu \geq \boxed{1.75} \\ H_a: \mu < 1.75 \end{cases}$$

$n = 36$, $\bar{x} = 1.65$

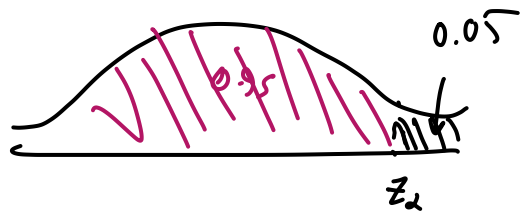
$\sigma = 0.12$

(b) $$Z_{\bar{x}} = \frac{\bar{x} - \mu_{H_0}}{\sigma/\sqrt{n}} = \frac{1.65 - 1.75}{0.12/\sqrt{36}} = -5$$

Critical value Method $\xrightarrow{\alpha = 0.05}$ left tailed test

left tailed test $\quad Z_{\bar{x}} < -Z_\alpha$

1.645



$|Z_{\bar{x}}| > Z_\alpha$

Correct decision Rule

-5

$Z_{\bar{x}} < -1.645 \longrightarrow$ Reject

$|Z_{\bar{x}}| > 1.645$

$|-5| > 1.645$

Reject

P-value?

P value: $P(Z < Z_{\bar{x}}) = P(Z < -5) < 0.001 < 0.05$

3. To determine the average price of hotel rooms in Atlanta, a sample of 49 hotels was selected and yielded an average price of hotel rooms being \$120. The population standard deviation was found to be \$16.

   a. Formulate the hypotheses to determine whether the average price of hotel rooms is significantly different from \$124.5.

   b. Calculate the test statistic.

   c. At 10% level of significance, use critical value approach to test the hypotheses. What is the conclusion?

   d. At 90% confidence, using the confidence interval approach to test the hypotheses. What is the conclusion?